Sentiment Analysis

Data Visualization

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Abstract

This project discusses the problem of sentiment analysis in Twitter, classifying tweets according to the sentiment expressed in them: positive, negative, or neutral. Twitter is an online microblogging and social networking platform which allows users to write short status updates about anything they want. It is a rapidly expanding service with over 206 million registered users.[1] - out of which 100 million are active users and half of them log on twitter on a daily basis - generating nearly 500 million tweets per day [2]. Analyzing the public sentiment is important for many applications, such as companies trying to find out the response of their newly launched products in the market, predicting political elections, and artists/singers trying to find out the response of public for their new songs or films. This project aims to develop a functional classifier in which a user will input the #tag to search the tweets, and our model will predict the sentiments on the tweets occurring in that #tag. Also, it will display a word cloud of top occurring words so that the user can have an idea about which are the most used words in each of the sentiments, i.e., positive, negative, and neutral.

Keywords*:* Sentiments, Twitter, Classification, #tag, Word Cloud.

Introduction

IoT focus on integrating 4 perspectives i:e People, Thing, Data, and Process. Artificial intelligence (AI) and machine learning may be used by IoT to make data collection procedures simpler and more dynamic. In this project, we have used Machine learning algorithms and Artificial intelligence to use the data and visualize the results so that it is understandable for normal persons.

This project is using the data from Twitter, One of the most used Microblogging social media platforms, in real-time and then giving us back the sentiments on the tweets which it gets from Twitter. Our app will show the Positive, Negative, and Natural sentiments on the Tweets. Along with this, it will also show the Word cloud for each sentiment so that we as a user have a better and clear understanding of the data (Tweets) which we get from Twitter.

For this project, we have used the Twitter API V2. To get the tweets in real-time from Twitter. This Application will ask the user to enter the #tag they want to search the sentiments for and also the number of tweets they want to search. After that our Application will show the text of tweets in one window along with the sentiments in the form of -1 0 1 in front of each tweet. And for making it easier to understand for users our Application will show the sentiments of all tweets in the form of Bar graph and Pie chart as well. It will also show the Word clouds for each sentiment so that users have a clear image of what are the most used words in the tweets they searched for.

This Application is will be very helpful for artists who just launched their project and they want to check the feedback of the public or for the company which launch a new product in the market and they want to get the sentiments of users.

Background

Many companies, for example, mobile phone companies do their market research on customers by feeding them “leaked” information so that they can analyze the feedback of users and they can make decisions based on those analyses. So without reading every tweet one by one they can perform sentiment analysis on the #tag at once and our Application will give them sentiments along with top occurring words in those tweets so that they have a clear image about the feedback and they will save a lot of time and effort. The same applies to the artists and other users who want to check the sentiments of the public against their product or their work (Song, Film trailer, etc.). It will be useful for everyone interested in checking the sentiments of the public on any product, Film, Song, or any event happening in the world by just entering the #tag and number of total tweets in our Application.

Method

To ensure that this project was to be successful and must be completed within the due time we made a project plan. We had divided our work into sub-tasks so that we can focus on one task at a time. We have chosen this project of data visualization because we have little experience of using the python language and we were intended on making an application which not only uses Machine learning models but also provides ease to its users in terms of time and effort. We have used the Twitter API for fetching the Tweets from Twitter as without that it is nearly impossible to fetch tweets from Twitter and even if someone tries to do so Twitter blocks their IP Address.

For the development of this application, we have used Visual Studio Code and we have used Python as the main programing language and we have used JavaScript, and HTML for the frontend, and made a simple and easy to use web-based GUI so that anyone can use the application with ease. Neither of us was good with JavaScript as a programing language so for that we have taken a look at a few tutorials on JavaScript and for showing charts we have used Chart.js library and for showing tables we have used Tabulator.

Design and Development

In this section, we will discuss the design of the application and how it has been developed. We will describe which tool and APIs we have used for obtaining the results.

Data Gathering

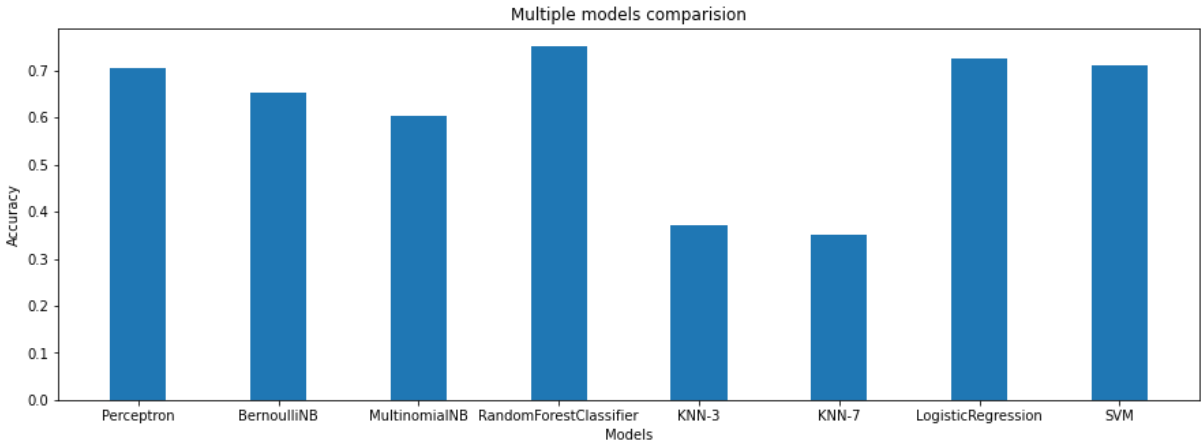
We needed the dataset of tweets in the form of CSV so that we can use that dataset to train our model for predicting the sentiments of tweets. We searched for the relevant data set on the internet and we found one from Kaggle. Our data set was already cleaned and annotated, Labels were available with the data set. Our data set contain two things, Text, and Category. In the text, there are Tweets and in the category, there is a sentiment of these tweets in the form of -1 0 1, -1 being Negative 0 being Neutral and 1 being Positive. It was a little biased data set as all of the tweets present in the dataset were Indian tweets and they are mainly about Indian politics. There are a total of 162980 Tweets in the CSV we used to train our model for sentiment analysis.

Model Training and Model Selecting

After finding the dataset move to the next step of model training and selecting the right model for our application. We split our dataset in two, for training our model we used 70% data, and for testing, we use 30% data. We first converted our dataset into vectorizer form and made a word count vectorizer(use the vectorizer to extract the features and use them to train our model).

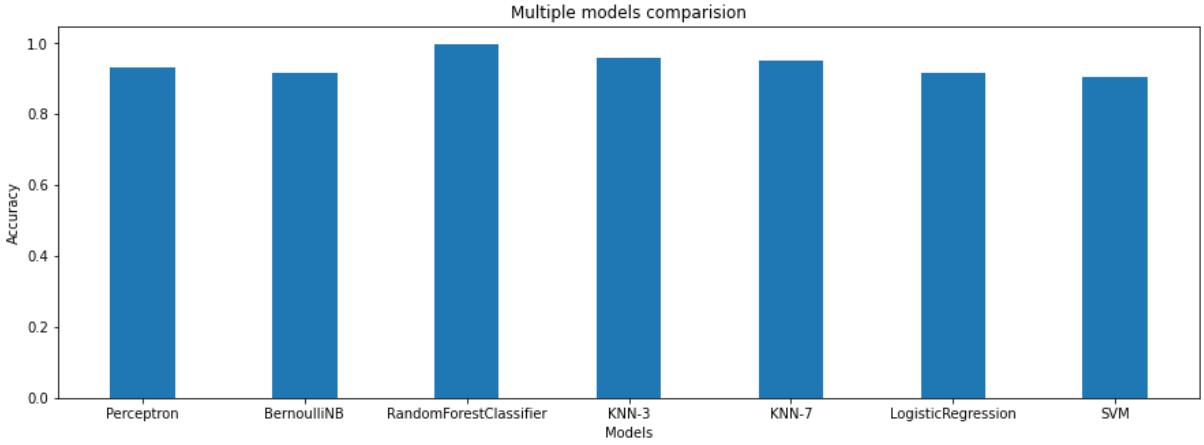
After making the vectorizer we applied machine learning classification models such as " Perceptron, BernoulliNB, KNN-3, KNN-7, MultinomialNB, Random Forest, SVM, and Logistic Regression". Out of all these models Random Forest was the best performing model and it was giving us an accuracy of 74.9% (0.749). We also kept "Precision = 77% (0.77), Recall = 73% (0.73), and f1-score = 73%(0.73)" in our consideration as well so that we should choose the best possible model for our application.

These were the results for all models we get:



After that for finding a sentiment using python's prebuilt libraries we used a library named Afinn which is used for calculating the Afinn_score and we used the "TextBlob" library to find the "Subjectivity" and "Polarity" of the next sentence. And when we used these two to find the sentiments of the tweets it gave us much higher Accuracy. Not only Accuracy but Recall, Precision, and Recall was improved significantly as well. But in this case, too Random Forest was the best performing Model with the Accuracy of 99%.

These were the results when we were using Afinn and TextBlob libraries:



Although they give us much higher accuracy we thought it will not be a good idea in terms of our learning to use everything prebuilt so we worked with a normal vectorizer and move further with it. We stored the best performing Model in our case it was Random Forest and vectorizer which we use to create features which was formed on Training set of our dataset(70% of total data) in a pickle format which we further used in our Flask Application.

Twitter API

After that, we requested permission to use the Twitter API. We needed the "Elevated" Access for our project. Getting permission to use that Elevated version of Twitter API was a bit hectic and time taking. We requested permission from two Twitter accounts, With one Account I requested Academic research access and with the other account, I requested Developer access. Twitter asked a series of questions regarding the project for which we are requesting access and several other details and then after a few days of analyzing our application they permitted us to use the API on developer access. After getting permission and keys to access the API from Twitter we wrote some APIs, In one

API we will give the Hashtag and count as an input and it access the tweets based on these two inputs, And another one is used for making charts for the sentiments present in the tweets and the last one is making a word cloud against each sentiment (Positive, Negative, and Neutral). The API for making charts works in a way that when we find the sentiments using the Model and the vectorizer we stored, we do predictive analysis and it gave us three classes in return which are Neutral, Negative, and Positive. Based on these we find the count for the number of classes which are positive, negative, and neutral which we finally used in our front-end for making Bar graph and Pie chart.

This Twitter API gives us the option of accessing the tweets on two bases, we can access the number of tweets that are most recent(Which we are doing in our Application) and the other option is to access the popular tweets. Previously Twitter does allow to access the tweets based on timeline and users do have the option of accessing the tweets by entering the time duration for which they want to access the tweets but recently they removed this option and now they only allow to access the tweets on only these two bases which I have mentioned above.

Word Cloud

After the prediction of sentiments is done we separated the tweets based on their sentiments, Positive, Negative, and Neutral and after that, the words which are present in these tweets after cleaning are used in making word cloud. We have used the “WordCloud” library which is an open-source library of python for making Word clouds. We have made separate word clouds for each sentiment.

Here are the sample word clouds for each sentiment when we tested this “WordCloud” library on our test data.

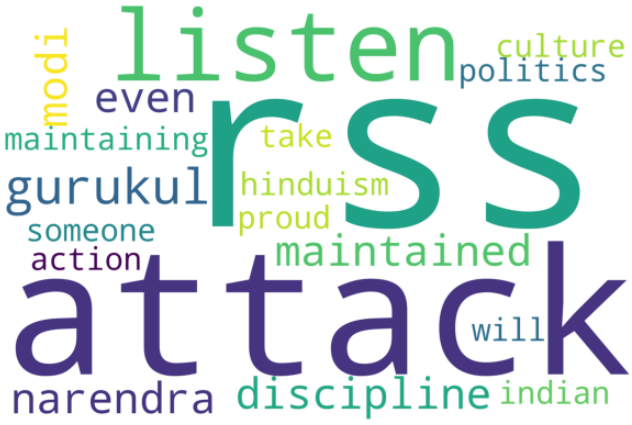
Neutral Sentiment Word cloud:



Positive Sentiment Word cloud:



Negative Sentiment Word cloud:



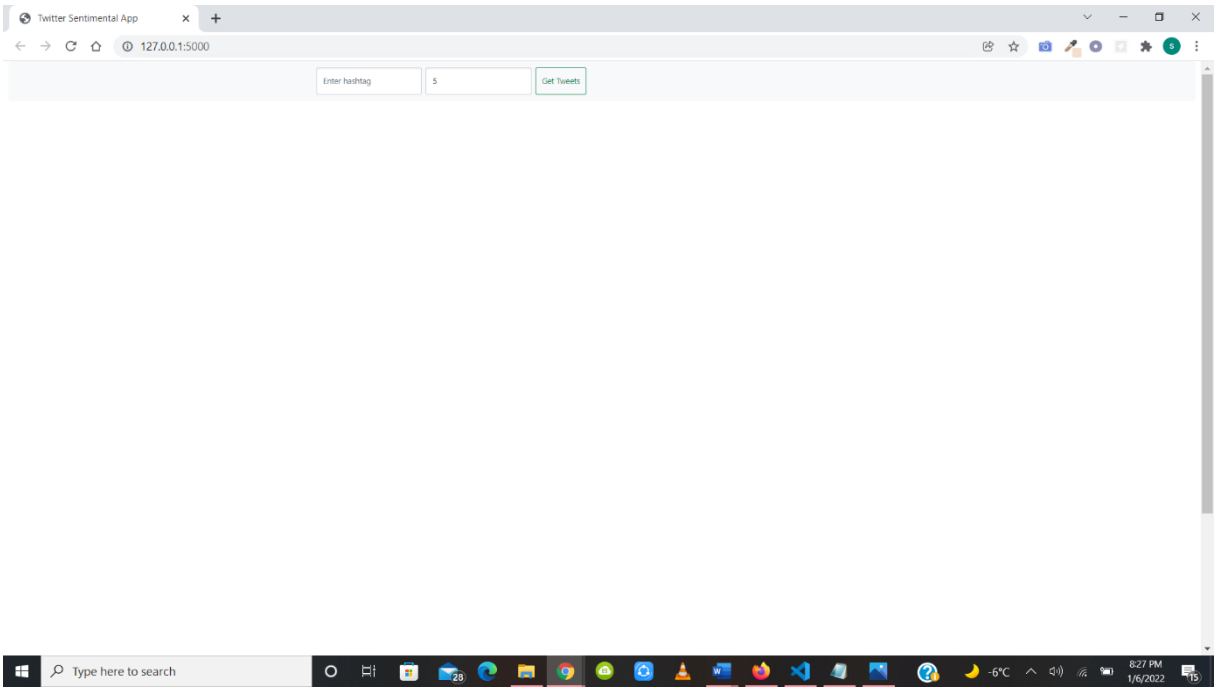
Frontend

We have used jQuery for communication between the Backend and Frontend of our application. Apart from this we have also Bootstrap for making our GUI responsive. We have used Chart.js for making charts of the sentiments, We are making Bar graph as well as Pie chart for the sentiments to make things as clear as possible for the users. We have also used Tabulator for making tables. These are the publicly available open-source JavaScript libraries. We have made a GUI using HTML and on the button “Get Tweets” when clicked send a request to our Flask API.

Results

We have achieved what we planned and promised in our Project plan and our First presentation. The accuracy of our application is ~75% which can be increased to 99% if we shift our application to pre-existing libraries like TextBlob. We have hosted our Application on Local Host.

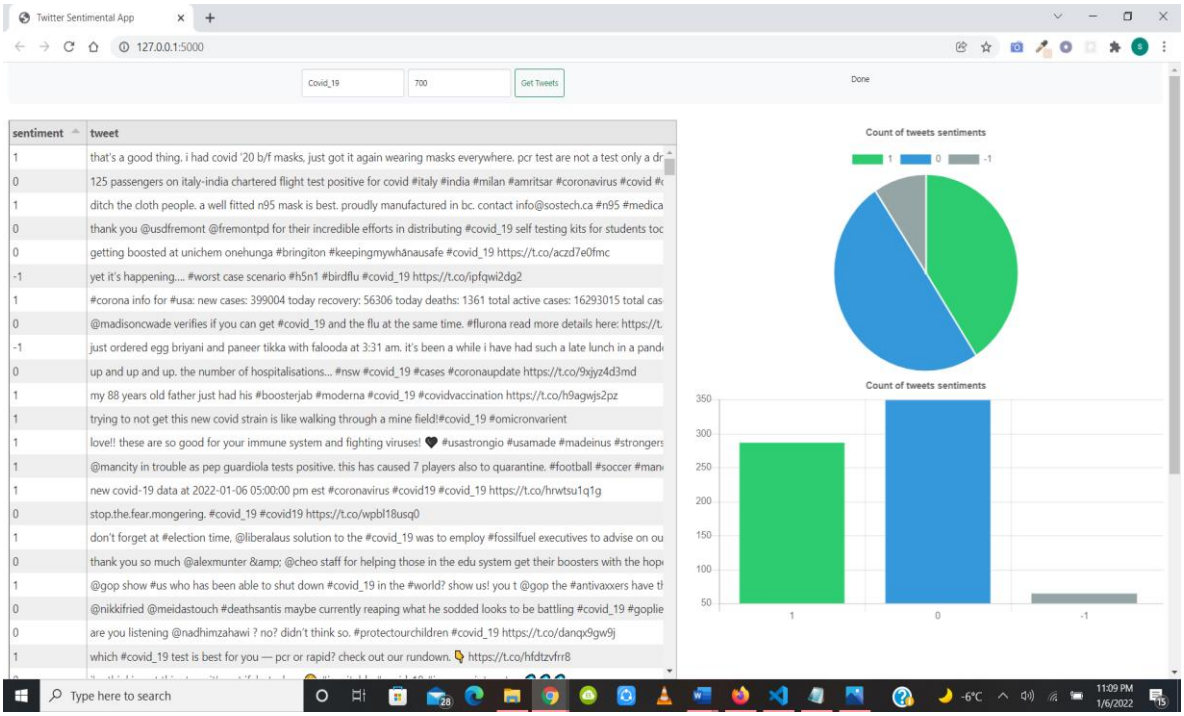
Here is the first page of our Application when it is in running form.



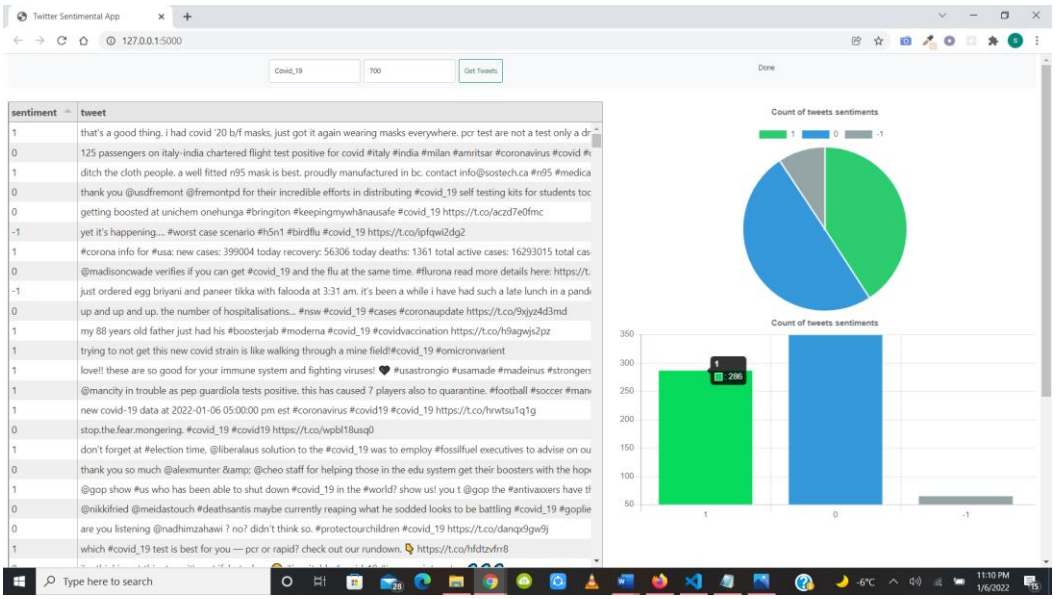
On this page user is required to input any hashtag they want, they can enter the hashtag with or without "#" as it will not be an issue as we have already kept this issue in mind while writing the code. So if any user wants to search for the Covid_19 Hashtag and if he/she enters "Covid_19" it will work the same as if he/she enters "#Covid_19" and our application will work perfectly fine in both of these situations. So after writing a Hashtag user will enter the total number of tweets he/she want to search. And will click on the "Get Tweet" Button. After pressing this button our application will show "Please wait for fetching and performing sentiment analysis... " right next to the button like this.



After some time (Depending on the number of tweets the user wants to fetch) new page will open with the results. Which will be like this.

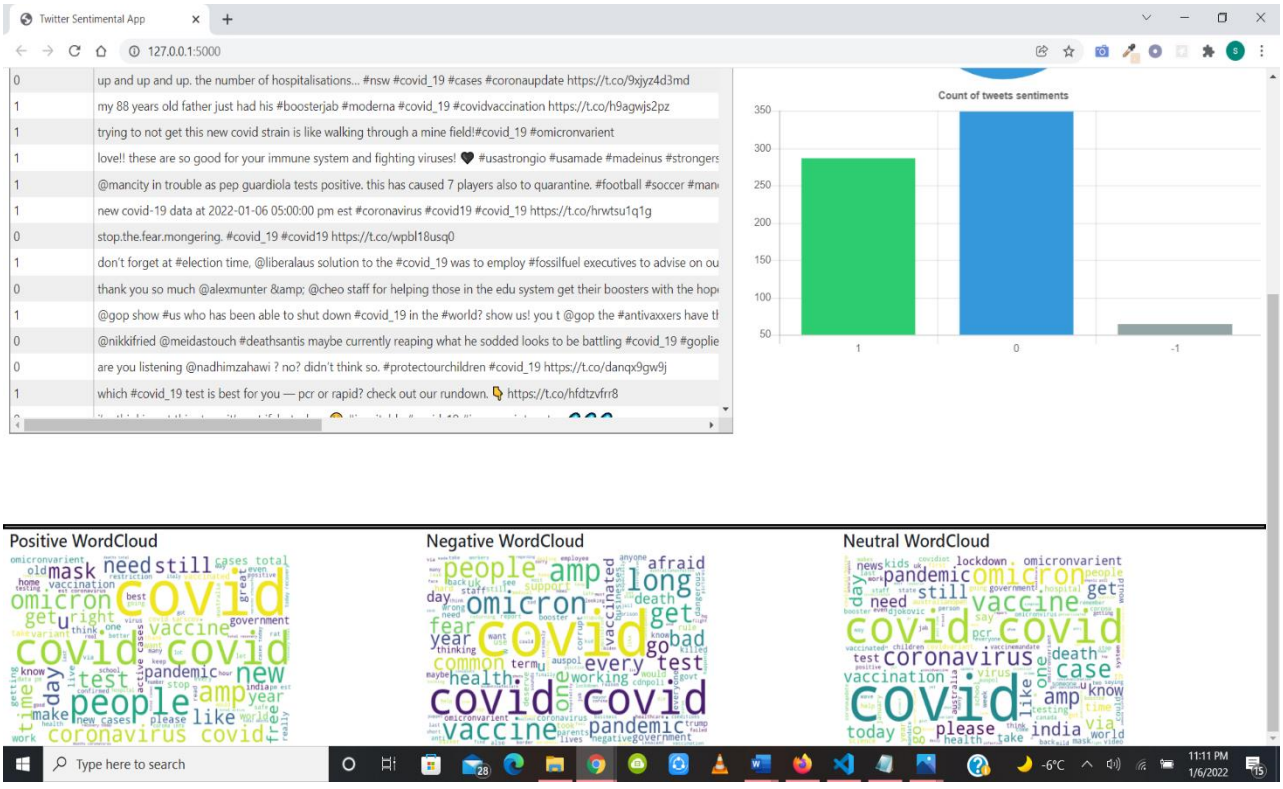


On the left side of the screen, we have shown the tweets which were fetched along with their sentiments on the extreme left column of that table. In this table, we are showing the tweets as it is without any cleaning and removing of stop words but on the back end before feeding these tweets to our model for performing the sentiments we are cleaning the data and removing the stop words from these tweets so that we can achieve maximum results. On the right side, we are showing the sentiments of all the tweets in Pie chart as well as Bar graph for a better understanding of users. If a user takes the cursor and places it over let's say positive bar graph it will also show the total number of tweets that have positive sentiments. It is shown below:



Apart from these, we are also showing a word cloud against each sentiment. So we are showing separate word clouds for tweets that are labeled as positive and separate for negative and Neutral. In each word cloud, we are showing the most occurring words so that the user has an idea that which are the most occurring words in the tweets and in which sentiments they are used. We have shown the word cloud at the end of the page below the tweets table and sentiments bar graph.

Here is the word cloud for the Hashtag “Covid_19”



Discussion

The reason for choosing this particular project is that both of us had a bachelor's in Computer Science and have little experience with python. Apart from this, we were also interested in learning about Natural Language Processing (NLP), and combining it with IoT is an exciting thing for us. Previously we have seen and even done some work on Sentiment analysis but all of that was done only using the existing data sets and just training the models using machine learning algorithm but in this, we moved to the next step and we are intended to make an application which benefits general users and public by enabling them to find the sentiments of any Hashtag they want. In this way, we are not restricted to the only dataset of tweets available to us but we are constantly fetching the data in the form of tweets from Twitter and whenever we fetch the data from Twitter it will get us new and fresh data of tweets.

So the reason for making this application is that it will provide ease to its users in terms of time-saving as well as effort saving in a way that if any user wants to check that the Hashtag which is now trending on Twitter is going in a positive direction or a negative direction in terms of the sentiments of the users tweeting about the Hashtag. So instead of wasting their time and effort by reading tweets and then analyzing it they can just simply put that Hashtag in our application and they can enter the count they want and our Application will give them the results in a matter of seconds that too with a lot more information which user will gather on its own like most occurring words in the tweets.

Future Work

As of now, we are using our Laptop as a server for hosting this Application because of the time constraint. But in the future, we can make this application public assessable by making Andriod and IOS Applications of our App and we can also buy a domain and hosting so that we can make a proper commercial use website for our Application.

We can also add some more features(If possible) like we can give users a location from where most numbers of tweets are posted and we can also add another feature on the first page of our Application which along with the Hashtag and Count user can also limit it search on the bases of location so that it will give them more accurate and more relevant data for their usage.

We can expand this model by adding an option of checking sentiments on the movie reviews by giving just the movie name. We can make the connection with IMDB to access reviews of the general public about a particular movie and then perform sentiment analysis on that. This will surely increase the usage of our application and it will provide ease to the users which are interested in watching movies as they will have an idea about the sentiments of other people which have already watched this movie.

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2: Total Number of tweets daily " https://www.internetlivestats.com/twitter-statistics/ "